

MV "Raphael"	INITIAL SURVEY		FINAL SURVEY	
	Inner roads 9.1		Wharf 212	
	18.04.2026 - 08:40 h		22.04.2026 - 19:15 h	
	0,20 m. waves		0,10 m. waves	
IMO NO. 1030806	FLAG: Liberian	BUILD: 2025	Wind force 02	Wind force 01
DRAUGHT SURVEY CERTIFICATE			Ebb tide, < 1 knot	Ebb tide, < 1 knot
PGUA/DS/123/26			Weather: Fine	Weather: Fine
DRAUGHT	INITIAL		FINAL	
Forward port	5,2900 m		12,8100 m	
Forward starboard	5,0800 m		12,7900 m	
Forward mean	5,1850 m		12,8000 m	
Stern correction	-0,1101 m		-0,0215 m	
Draught forward (corrected to fore perpendicular)	5,0749 m		12,7785 m	
Midship port	6,2700 m		12,9400 m	
Midship starboard	5,8800 m		12,8900 m	
Midship mean	6,0750 m		12,9150 m	
Midship correction	-0,0304 m		-0,0176 m	
Draught midship (corrected to midship perpendicular)	6,0446 m		12,8974 m	
Aft port	7,3500 m		12,9100 m	
Aft starboard	7,3400 m		12,8800 m	
Aft mean	7,3450 m		12,8950 m	
Stern correction	-0,0361 m		-0,0096 m	
Draught after (corrected to after perpendicular)	7,3089 m		12,8854 m	
Kell correction	0,0170 m		0,0170 m	
Waterplane coefficient correction for hull deflexion	0,0157 m		-0,0096 m	
Means of means corrected for deflexion:	6,0971 m		12,8715 m	
True trim	BY STERN	2,23 m	BY STERN	0,11 m
Heel (angle & in cm)	P 001 °	39,00 cm	P 000 °	5,00 cm
Hull deflexion:	HOGGING	14,73 cm	SAGGING	6,55 cm
DISPLACEMENTS	INITIAL		FINAL	
Displacement in salt water	37.352,6 t		83.865,5 t	
First trim correction	-321,9 t		12,5 t	
Second trim correction	49,6 t		0,0 t	
Displacement corrected for trim	37.080,3 t		83.878,1 t	
Water density	1,01800		1,01900	
Displacement corrected for density	36.827,0 t		83.387,1 t	
Deductibles (fresh water, oils, etc.)	23.041,5 t		4.853,3 t	
Displacement corrected for deductibles	13.785,5 t		78.533,8 t	
Light Ship	13.515,9 t			
Calculated constant	302,9 t			
Anchor + chain	33,3 t		0,0 t	
Total cargo loaded on board as per draught survey			64715,0 t	
Loading terminals: CO, .	TPC	65,1 t/cm	TPC	71,0 t/cm
Cargo: Soybean meal in bulk	MTC +0.5	967,6 t*cm	MTC +0.5	1230,9 t*cm
Paranagua, Br - Brazil. April 22nd, 2026	MTC -0.5	922,8 t*cm	MTC -0.5	1220,9 t*cm
Total cargo loaded on board as per shore scales			64.765,300 t	
Total cargo loaded on board as per draught survey			64.715,034 t	
Shortage	-50,266 t		-0,08 %	

Remarks: 1) Due to huge hull deflexion in hogging on initial draft survey and sagging on final draft survey, it was applied the waterplane area coefficient correction for the deflexion formula as per UN/ECE/Energy/19/1992. 2) According to the Master/Chief Officer usual constant would amount to 300 t.

Standard Brazil Marine Surveys & Services Ltda.

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Licensed Marine Officer / Marine Surveyor / 375460MM

Master/Ch.Off: 64.630,748 t -134,552 t -0,21 %

Customs : 64.768,421 t 3,121 t 0,00 %

Owners P&I: 64.763,700 t -1,600 t 0,00 %

Master/Ch.Off.